CS 229 HW2 EC Report

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Additional Calibration Methods for the Emoji Hunter Task

Problem & Approach

In our original analysis, the base model achieved 78.0% accuracy with ECE = 0.1278. I explored the following methods as well.

- **Platt Scaling** (post-hoc): Fits a logistic regression on the model's raw logits using a held-out validation set, learning both a slope and bias.
- **Weight Decay** (train-time): Adds an L2 penalty $\lambda\|w\|^2$ to the training loss, encouraging smaller weights. We tested $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}\}$ with Adam and selected the best ECE.
- **Focal Loss** (train-time): Modifies cross-entropy with a modulating factor $\text{FL}(p_t) = -(1 - p_t)^\gamma \log(p_t)$ that downweights easy examples. We tested $\gamma \in \{1.0, 2.0, 3.0\}$.
- **Mixup** (train-time): Trains on combinations of input pairs $\tilde{x} = \lambda x_i + (1 - \lambda)x_j$ with blended labels, where $\lambda \sim \text{Beta}(\alpha, \alpha)$. This forces intermediate confidences. We tested $\alpha \in \{0.1, 0.2, 0.4\}$.

Results

Type	Method	Accuracy	ECE ↓
—	Base Model	78.00%	0.1278
Post-hoc	Temp Scaling ($T = 3.0$)	78.00%	0.0373
Inference	MC Dropout (30 passes)	77.75%	0.1022
Train-time	Label Smoothing ($\alpha = 0.1$)	78.00%	0.0397
Post-hoc	Platt Scaling	77.25%	0.0525
Train-time	Weight Decay ($\lambda = 10^{-2}$)	77.75%	0.1066
Train-time	Focal Loss ($\gamma = 2.0$)	78.75%	**0.0363**
Train-time	Mixup ($\alpha = 0.4$)	76.75%	0.0805

Table 1: Calibration comparison across all methods. ECE = Expected Calibration Error.

All of the evaluated methods improved calibration over the base model (ECE = 0.1278). The best overall method was Focal Loss ($\gamma = 2.0$), achieving an ECE of 0.0363, followed by Temperature Scaling (0.0373).

- **Train-time methods: Focal Loss** ($\gamma = 2.0$) was the most effective (ECE = 0.0363). It did calibration better than Label Smoothing, Mixup, Weight Decay in the train time methods.
- **Post-hoc (Test time methods): Temperature Scaling** ($T = 3.0$) performed best (ECE = 0.0373). It outperformed Platt Scaling, likely because Platt Scaling's additional parameters were prone to overfitting on the small validation set.
- **Inference time methods: MC Dropout** (30 passes) provided only a modest improvement (ECE = 0.1022).